



July 14, 2025

HD Hyundai Marine Solution (443060)

2Q25 preview: A steady stream of growth

2Q25 OP in line; revenue trimmed on lower non-core bunkering sales

We forecast 2Q25 consolidated revenue at KRW499.4bn (+14.1% yoy, +2.8% qoq) and OP at KRW88.7bn (+24.9% yoy, +6.8% qoq; OPM 17.8%), broadly in line with consensus. We revised down our revenue and OP estimates by 6.5% and 6.9%, respectively, for two reasons. First, we lowered our forecast for the bunkering segment revenue by 7%, reflecting a 9.8% and 7.5% drop in HSFO bunker prices in Korea and Singapore, respectively. Given the segment contributes only 0.6% to OP and is non-core, the impact on earnings is immaterial.

Revenue adjustment to AM solutions on refined ship count and seasonality

Second, we cut our AM solutions revenue forecast by 7.3%, adjusting the number of newly serviced vessels from 633 to 570 (-8.8%) based on updated engine delivery volumes at HD Hyundai Heavy Industries (1,300 units annually: 300 large, 1,000 medium). The revised figure reflects both internal and external customers. AM revenue is allocated quarterly from annual projections; we reduced the 2Q25 allocation ratio by 1%p to 24.5% and raised the 3Q25 ratio to reflect seasonality.

TP raised 4%; growth intact, with further upside from retrofit orders

The long-term trajectory of simultaneous price and volume growth in AM services remains unchanged. While we continue to refine our quarterly modeling, our confidence in full-year earnings remains intact. We reiterate our BUY rating and raise our TP by 4.2% to KRW250,000. The TP is derived by dividing our estimated fair market value of KRW11.3tn by the number of outstanding shares. The fair value is based on 2026F NP of KRW351.1bn and a target PE of 32.1x, which reflects a 30% premium to the average 2026 PE of three peers, taking into account the company's three-year earnings CAGR. If green retrofit solution orders are secured in 2H25, they could offer further upside potential to both earnings and valuation.

Yr to	Sales	OP	NP	EPS	% chg	EBITDA	PE E/EBITDA	PB	ROE	DY
Dec	(W bn)	(W bn)	(W bn)	(KRW)	(YoY)	(W bn)	(x)	(x)	(%)	(%)
2023A	1,430	201	151	3,778	44.0	212	NM	NM	NM	71.6
2024A	1,745	272	228	5,304	40.4	284	30.4	23.7	9.4	45.0
2025F	2,040	363	285	6,362	20.0	376	29.6	21.1	9.8	34.8
2026F	2,162	438	351	7,832	23.1	452	24.1	17.5	8.6	38.0
2027F	2,403	537	435	9,705	23.9	552	19.4	14.2	7.5	41.4

Note: Net profit and EPS are based on figures attributed to controlling interest

12M rating **BUY (Maintain)**

12M TP **W250,000** from W240,000
Up/Downside **+33%**

KIS ESG Indicator

Composite	E	S	G
NA	NA	NA	NA

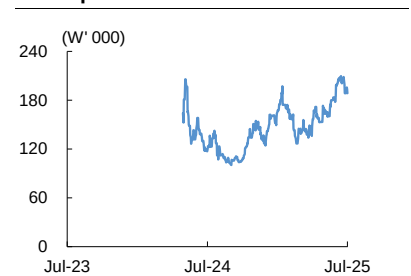
Stock data

KOSPI (Jul 11, pt)	3,176
Stock price (Jul 11, KRW)	188,600
Market cap (USD mn)	6,147
Shares outstanding (mn)	45
52-Week high/low (KRW)	209,500/100,100
6M avg. daily turnover (USD mn)	26.5
Free float / Foreign ownership (%)	31.0/29.8
Major shareholders (%)	
HD HYUNDAI CO.,LTD. and 3 others	55.3
Global Vessel Solutions, L.P.	10.0

Performance

	1M	6M	12M
Absolute (%)	(5.0)	12.6	59.4
Relative to KOSPI (%p)	(14.3)	(13.6)	49.6

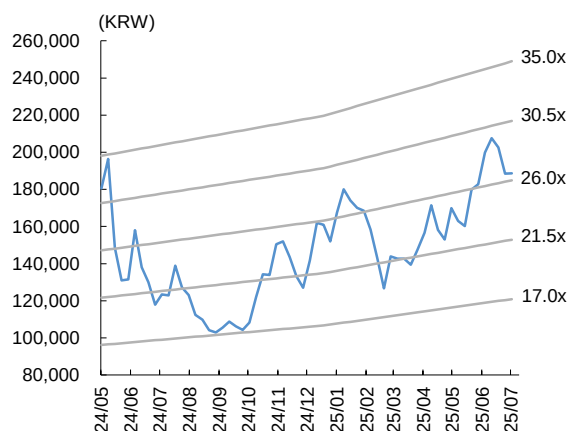
Stock price



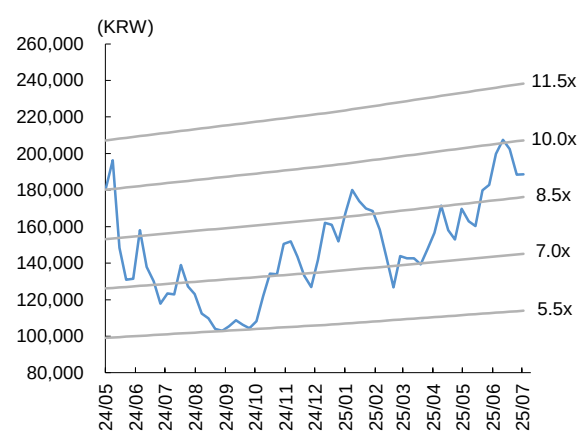
Source: FnGuide

Kyungtae Kang

kyungtae.kang@koreainvestment.com

Figure 1. 12MF PE band

Note: As of July 11, 2025
Source: Quantwise, Korea Investment Securities

Figure 2. 12MF PB band

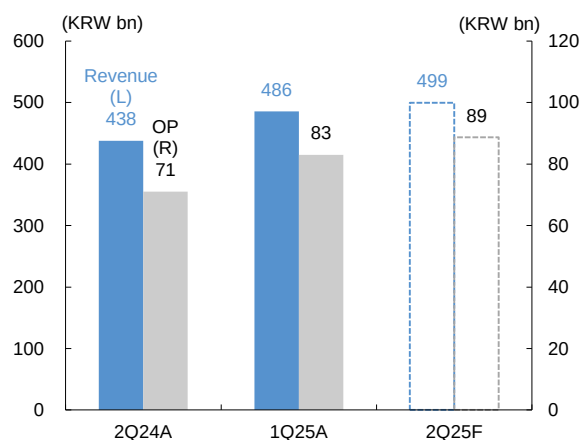
Note: As of July 11, 2025
Source: Quantwise, Korea Investment Securities

Table 1. 2Q25 earnings

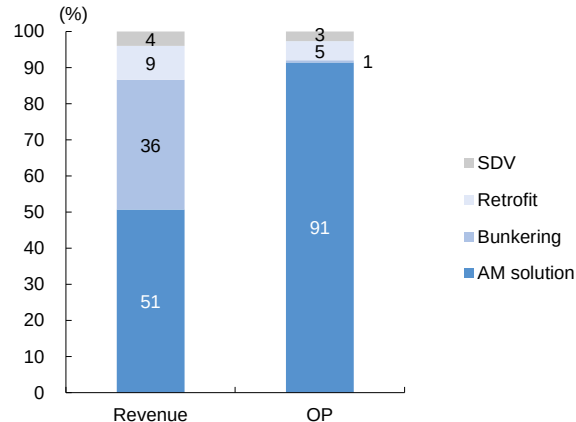
(KRW bn, %)

	2Q25					qoq		yoy	
	Preview	Previous	Diff.	Consensus	Difference	1Q25	Diff.	2Q24	Diff.
Revenue	499	534	(6.5)	512	(2.5)	486	2.8	438	14.1
- AM solution	253	273	(7.3)			216	17.2	206	22.9
- Bunkering	179	193	(7.0)			191	(6.3)	178	0.9
- Retrofit	47	48	(2.2)			58	(19.6)	35	35.4
- SDV	20	20	0.0			20	1.1	20	2.6
OP	89	95	(6.9)	86	2.6	83	6.8	71	24.9
- AM solution	81	87	(7.3)			75	8.4	66	23.2
- Bunkering	1	1	(7.0)			0	1,079.7	0	13.1
- Retrofit	5	5	(2.2)			6	(19.6)	2	93.4
- SDV	2	2	0.0			2	1.1	2	2.6
OPM	17.8%	17.8%	(0.1%p)	16.9%	0.9%p	17.1%	0.7%p	16.2%	1.5%p
NP	70	75	(6.5)	69	0.9	63	10.5	57	23.4

Source: Dataguide, Korea Investment Securities

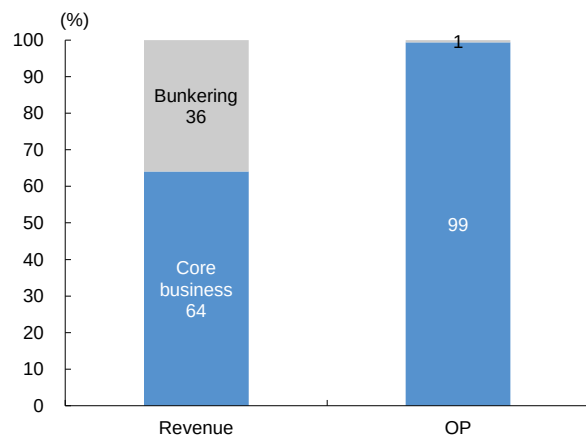
Figure 3. yoy and qoq comparison of 2Q25 results

Note: Consolidated
Source: DART, Korea Investment Securities

Figure 4. OP contribution by business segment

Note: Consolidated
Source: Korea Investment Securities

Figure 5. Profit contribution: core vs. non-core (bunkering)



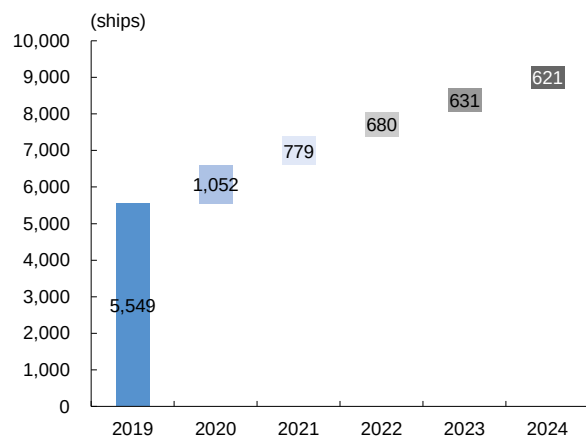
Note: Consolidated
Source: Korea Investment Securities

Figure 6. LNG-FSU retrofit project completed on July 10, 2025



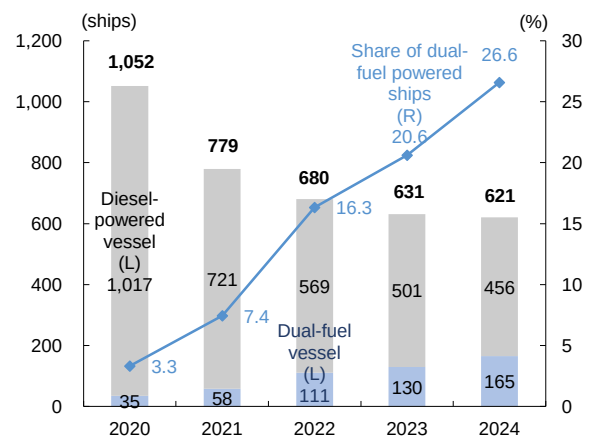
Note: Contracted in August 2024 for USD 30mn retrofit of the Bilbao Knutsen
Source: Money Today

Figure 7. Cumulative number of ships served with AM services



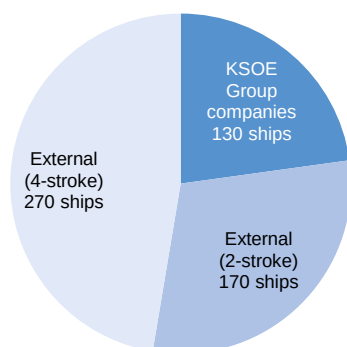
Note: Bars and labels represent net additions since 2019
Source: HDHSM, Korea Investment Securities

Figure 8. Net increase in AM-serviced ships by engine type



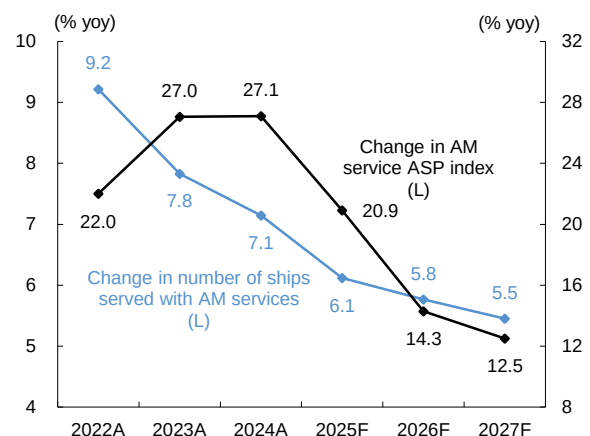
Source: HDHSM, Korea Investment Securities

Figure 9. Annual increase in ships served with AM services projected at 570



Note: Based on 1,300 engines delivered annually by HDHHL Engine & Machinery
Source: HDHMS, HDHHL, Korea Investment Securities

Figure 10. AM solutions revenue forecast: change in P and Q



Source: HDHSM, Korea Investment Securities

Table 2. Quarterly earnings

(KRW bn, % yoy, % qoq, %)

	1Q25A	2Q25F	3Q25F	4Q25F	1Q26F	2Q26F	3Q26F	4Q26F	2024A	2025F	2026F	2027F
Revenue	486	499	540	515	525	545	538	554	1,745	2,040	2,162	2,403
Core business	294	320	367	340	350	375	368	381	1,040	1,321	1,474	1,703
AM solution	216	253	275	266	286	299	324	336	806	1,010	1,245	1,509
Engine	157	167	181	177	190	198	215	223	532	683	825	979
Non-engine	28	34	37	35	37	38	41	43	116	134	159	191
Land power generation	9	26	28	27	29	31	33	34	70	90	127	171
Smartcare (warranty)	22	26	29	27	31	32	35	36	87	103	133	168
Eco-friendly solution (retrofit)	58	47	72	54	39	52	20	20	164	231	132	82
BWTS	0	0	0	0	0	0	0	0	0	0	0	0
EGCS	0	0	0	0	0	0	0	0	0	0	0	0
Green solution	34	47	72	54	39	52	20	20	0	207	132	82
Gas solution	34	47	14	27	14	27	0	0	0	122	40	0
CO2 solution	0	0	58	26	26	26	20	20	0	85	92	82
Digital solution (SDV)	20	20	20	20	24	24	24	24	71	80	97	112
Non-core business	191	179	173	175	175	170	170	173	706	719	688	700
Bunkering	191	179	173	175	175	170	170	173	706	719	688	700
OP	83	89	98	93	102	107	112	116	272	363	438	537
Core business	83	88	98	93	101	107	112	116	270	362	436	535
AM solution	75	81	88	85	95	99	107	111	250	329	411	513
Eco-friendly solution (Retrofit)	6	5	7	5	4	5	2	2	11	23	13	8
Digital solution (SDV)	2	2	2	2	3	3	3	3	8	10	12	13
Non-core business	0	1	1	1	1	1	1	1	2	2	2	2
Bunkering	0	1	1	1	1	1	1	1	2	2	2	2
EBT	84	93	103	99	108	114	120	125	297	380	468	580
Controlling-int. NP	63	70	77	74	81	86	90	94	228	285	351	435
Growth (% yoy)												
Revenue	26.8	14.1	17.0	11.2	8.1	9.2	(0.3)	7.5	22.0	16.9	6.0	11.1
Core business	35.6	23.0	31.0	20.2	19.0	17.2	0.3	12.0	24.1	27.0	11.6	15.5
AM solution	27.6	22.9	25.4	25.9	32.5	18.0	17.8	26.4	32.7	25.4	23.3	21.2
Eco-friendly solution (retrofit)	92.4	35.4	64.6	(1.9)	(32.3)	11.4	(71.9)	(62.1)	(3.0)	41.3	(42.7)	(37.9)
Digital solution (SDV)	13.8	2.6	17.0	21.1	22.1	20.9	20.9	19.6	13.5	13.2	20.9	15.8
Non-core business	15.2	0.9	(4.7)	(2.9)	(8.7)	(5.2)	(1.5)	(1.2)	18.5	1.9	(4.3)	1.8
Bunkering	15.2	0.9	(4.7)	(2.9)	(8.7)	(5.2)	(1.5)	(1.2)	18.5	1.9	(4.3)	1.8
OP	61.2	24.9	17.7	41.8	22.7	21.0	14.4	24.6	34.9	33.7	20.5	22.6
Core business	62.6	25.0	17.7	42.2	22.2	21.1	14.5	24.7	35.4	34.0	20.5	22.7
Non-core business	(89.7)	13.1	10.2	(4.9)	1,049.4	(5.2)	(1.5)	(1.2)	(11.1)	(16.3)	26.8	1.8
Growth (% qoq)												
Revenue	4.8	2.8	8.1	(4.5)	1.8	3.9	(1.3)	2.9				
Core business	4.0	8.8	14.7	(7.4)	3.0	7.2	(1.9)	3.4				
AM solution	2.3	17.2	8.6	(3.2)	7.7	4.3	8.3	3.8				
Eco-friendly solution (retrofit)	6.4	(19.6)	53.9	(25.4)	(26.6)	32.2	(61.2)	0.7				
Digital solution (SDV)	18.6	1.1	0.0	1.1	19.6	0.0	0.0	0.0				
Non-core business	6.0	(6.3)	(3.7)	1.5	(0.4)	(2.7)	0.1	1.9				
Bunkering	6.0	(6.3)	(3.7)	1.5	(0.4)	(2.7)	0.1	1.9				
OP	26.0	6.8	10.7	(4.7)	9.0	5.3	4.7	3.7				
Core business	27.0	6.2	10.7	(4.8)	9.1	5.3	4.7	3.7				
Non-core business	(91.8)	1,079.7	(3.7)	1.5	(0.4)	(2.7)	0.1	1.9				
Key metrics												
OPM	17.1	17.8	18.2	18.1	19.4	19.7	20.9	21.0	15.6	17.8	20.2	22.3
Core business	28.2	27.5	26.6	27.3	29.0	28.4	30.4	30.4	25.9	27.4	29.6	31.4
Non-core business	0.0	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.3	0.3
AM solution	34.6	32.0	32.0	32.0	33.0	33.0	33.0	33.0	31.0	32.6	33.0	34.0
Eco-friendly solution (retrofit)	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0	7.0	10.0	10.0	10.0
Digital solution (SDV)	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0
Bunkering	0.0	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.3	0.3

Source: DART, company data, Korea Investment Securities

Table 3. Annual earnings forecast revisions

(KRW bn, KRW/share, %, KRW/USD, pt)

	Revised			Previous			Chg.		
	2025F	2026F	2027F	2025F	2026F	2027F	2025F	2026F	2027F
Sales	2,040	2,162	2,403	2,151	2,328	2,567	(5.2%)	(7.1%)	(6.4%)
OP	363	438	537	384	472	580	(5.5%)	(7.3%)	(7.5%)
OPM	17.8%	20.2%	22.3%	17.9%	20.3%	22.6%	(0.1%p)	(0.0%p)	(0.3%p)
EBIT	380	468	580	401	503	625	(5.2%)	(7.0%)	(7.2%)
NP	285	351	435	301	377	469	(5.2%)	(7.0%)	(7.2%)
EPS (KRW)	6,362	7,832	9,705	6,712	8,417	10,456	(5.2%)	(7.0%)	(7.2%)
BPS (KRW)	19,336	21,886	25,009	19,486	22,221	25,595	(0.8%)	(1.5%)	(2.3%)
ROE (%)	34.8%	38.0%	41.4%	36.5%	40.4%	43.7%	(1.8%p)	(2.4%p)	(2.3%p)

Source: Korea Investment Securities

Table 4. Valuation comparison of MRO peers

Company		Wärtsilä Corporation (WRT1V FH)	Alfa Laval (ALFA SS)	Kongsberg Gruppen (KOG NO)	HD Hyundai Marine Solution (443060 KS)
Share price (local currency)		20.1	423.2	326.1	193,400
Market cap (KRW bn)		19,063	25,207	39,032	8,670
PE	2024A	20.1	25.9	43.9	30.4
(x)	2025F	20.7	20.5	39.0	29.7
	2026F	19.0	19.2	36.0	23.9
	2027F	17.8	13.9	31.3	19.6
PB	2024A	4.0	4.6	12.1	9.4
(x)	2025F	4.2	3.7	12.9	9.9
	2026F	3.8	3.4	11.2	8.3
	2027F	3.5	3.0	9.8	7.0
ROE	2024A	11.1	15.4	27.0	23.9
(%)	2025F	11.9	12.6	26.8	21.6
	2026F	11.1	12.0	24.0	17.3
	2027F	10.6	9.5	21.0	14.2
Revenue growth	2024A	21.2	18.7	29.6	45.0
(% yoy)	2025F	21.0	19.8	34.3	35.2
	2026F	20.8	18.6	33.0	38.3
	2027F	19.8	17.7	33.2	39.6
OP growth	2024A	7.2	5.3	20.3	22.0
(% yoy)	2025F	10.4	4.4	21.4	18.3
	2026F	5.6	4.5	16.6	16.7
	2027F	4.6	3.9	13.5	15.7
OPM	2024A	79.1	13.0	41.5	34.9
(%)	2025F	11.3	14.7	34.7	30.7
	2026F	9.0	5.7	10.5	25.0
	2027F	6.2	4.2	16.2	23.0
NP growth	2024A	10.9	15.5	13.3	15.6
(% yoy)	2025F	11.0	17.1	14.8	17.2
	2026F	11.4	17.3	14.0	18.4
	2027F	11.5	17.3	14.3	19.6
NPM	2024A	95.0	16.8	38.1	50.8
(%)	2025F	12.5	17.2	34.1	24.4
	2026F	9.3	5.5	16.0	24.4
	2027F	6.2	39.0	15.2	22.2

Note: July 10 close

Source: Bloomberg, Korea Investment Securities

Table 5. TP raised 4.2% to KRW250,000

			Unit	Value	Note
2026F NP	(a)		KRW bn	351	52-week weighted
Target PE	(b)	Bloomberg	X	24.7	Three peers' 2026F PE average
Discount or premium	(c)	Peer study	%	30.0	
Target PE	(d)	$(d) = (b) \times (1 + (c))$		32.1	
EV	(e)	$(e) = (a) \times (d)$	KRW bn	11,275	
Outstanding shares	(f)	$(f) = (g) - (h)$	'000 shares	44,829	
No. of shares issued	(g)			44,829	
Treasury shares	(h)				
TP	(i)	$(i) = (e) / (f)$	KRW/share	250,000	
Current price	(j)		KRW/share	188,600	July 11 close
Upside	(k)		%	32.6	

Note: Controlling-int. NP

Source: Prospectus, Bloomberg, Korea Investment Securities

Income statement

(W bn)

	2023A	2024A	2025F	2026F	2027F
Sales	1,430	1,745	2,040	2,162	2,403
COGS	1,158	1,383	1,574	1,616	1,746
Gross profit	273	362	466	546	657
SG&A expenses	71	90	103	108	120
Operating profit	201	272	363	438	537
OP margin (%)	14.1	15.6	17.8	20.2	22.3
EBITDA	212	284	376	452	552
EBITDA margin (%)	14.8	16.2	18.4	20.9	22.9
Non-operating profit	0	25	17	30	43
Financial income	41	67	32	31	44
Financial expenses	41	39	14	1	1
Other non-operating profit	(1)	(3)	(1)	0	0
Gains (Losses) in associates, subsidiaries and JV	0	(0)	(0)	0	0
Earnings before tax	201	297	380	468	580
Income taxes	50	69	95	117	145
Consolidated net profit	151	228	285	351	435
Controlling-int. NP	151	228	285	351	435
Controlling-int. NP margin (%)	10.6	13.1	14.0	16.2	18.1
Growth (%)					
Sales growth	7.2	22.0	16.9	6.0	11.1
OP growth	41.9	34.9	33.7	20.5	22.6
Controlling-int. NP growth	44.0	50.8	25.1	23.1	23.9
EPS growth	44.0	40.4	20.0	23.1	23.9
EBITDA growth	41.6	34.1	32.6	20.1	22.1

Cash flow

(W bn)

	2023A	2024A	2025F	2026F	2027F
C/F from operations	73	247	294	291	384
Net profit	151	228	285	351	435
Depreciation	10	12	13	14	15
Amortization	0	0	0	0	0
Net increase in W/C	(97)	(6)	(5)	(74)	(67)
Others	9	13	1	0	1
C/F from investing	5	(196)	(45)	(26)	(40)
Capex	(3)	(10)	(21)	(22)	(23)
Decrease in fixed assets	0	0	0	0	0
Increase in investment	6	(186)	9	10	9
Net increase in intangible assets	0	(0)	(1)	(0)	(1)
Others	2	0	(32)	(14)	(25)
C/F from financing	(44)	149	(199)	(244)	(302)
Increase in equity	0	386	0	0	0
Increase in debt	(6)	(7)	3	3	3
Dividends	(90)	(100)	(141)	(202)	(247)
Others	52	(130)	(61)	(45)	(58)
C/F from others	(2)	3	0	0	0
Increase in cash	32	203	50	21	41
FCF	75	199	258	245	327

Note: 1) K-IFRS (consolidated financial statements)

Balance sheet

(W bn)

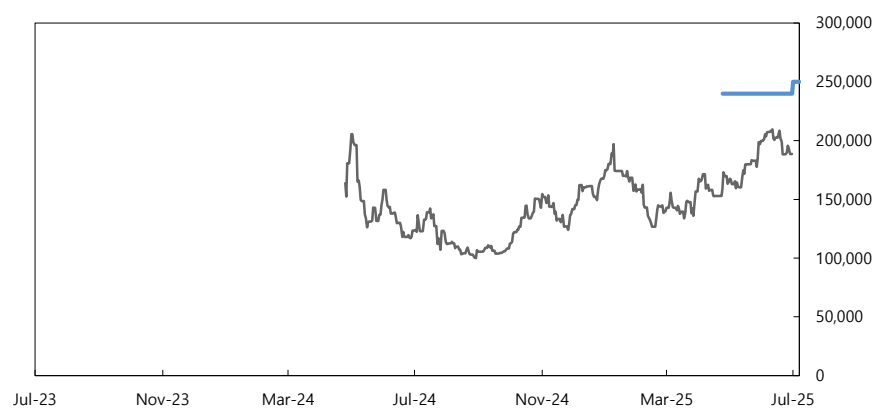
	2023A	2024A	2025F	2026F	2027F
Total assets	657	1,152	1,352	1,440	1,604
Current assets	597	1,076	1,257	1,332	1,481
Cash & cash equivalents	96	298	349	370	411
Accounts & other receivables	211	263	307	326	362
Inventory	238	273	320	339	377
Non-current assets	59	77	95	107	123
Investment assets	0	3	3	4	4
Tangible assets	11	20	28	36	45
Intangible assets	4	5	5	6	6
Total liabilities	417	379	485	458	483
Current liabilities	388	343	444	415	436
Accounts & other payables	146	184	215	228	253
ST debt & bonds	128	0	0	0	0
Current portion of LT debt	0	0	0	0	0
Non-current liabilities	29	36	41	43	47
Debentures	0	0	0	0	0
LT debt & financial liabilities	12	13	14	14	15
Shareholders' equity	240	774	867	981	1,121
Controlling interest	240	774	867	981	1,121
Capital stock	20	22	22	22	22
Capital surplus	20	385	385	385	385
Other reserves	1	30	30	30	30
Retained earnings	198	324	407	512	642
Minority interest	0	0	0	0	0
Net debt	50	(461)	(539)	(570)	(633)

Key financial data

	2023A	2024A	2025F	2026F	2027F
Per-share data (KRW)					
EPS	3,778	5,304	6,362	7,832	9,705
BPS	5,999	17,255	19,336	21,886	25,009
DPS	2,500	3,150	4,500	5,500	6,800
Profitability (%)					
ROA	25.7	25.2	22.8	25.2	28.6
ROE	71.6	45.0	34.8	38.0	41.4
Dividend yield	NM	2.0	2.4	2.9	3.6
Dividend payout ratio	66.2	61.9	70.7	70.2	70.1
Stability					
Debt/equity (x)	173.6	49.0	56.0	46.7	43.1
Int.-bearing debt/equity (%)	60.8	2.7	2.7	2.7	2.6
Interest coverage (x)	32.7	59.8	299.2	318.6	349.8
Net debt/EBITDA (x)	0.2	(1.6)	(1.4)	(1.3)	(1.1)
Valuation (x)					
PE	NM	30.4	29.6	24.1	19.4
PE peak	NM	39.1	34.0	27.6	22.3
PE trough	NM	18.8	19.8	16.1	13.0
PB	NM	9.4	9.8	8.6	7.5
PB peak	NM	12.0	11.2	9.9	8.6
PB trough	NM	5.8	6.5	5.8	5.0
PS	NM	4.0	4.1	3.9	3.5
EV/EBITDA	NM	23.7	21.1	17.5	14.2

Changes to recommendation and price target

Company (Code)	Date	Recommendation	Price target	% vs. avg. price	% vs. high (low)
HDHMS (443060)	07-05-25	BUY	W240,000	-22.5	-12.7
	14-07-25	BUY	W250,000	-	-



■ **Guide to Korea Investment & Securities Co., Ltd. stock ratings based on 12-month forward share price performance relative to the market index**

- BUY: Expected to outperform the market index by 15%p or more
- Hold: Expected to underperform or outperform the market index by less than 15%p
- Underweight: Expected to underperform the market index by 15%p or more
- Korea Investment & Securities does not offer target prices for stocks with Hold or Underweight ratings.

■ **Guide to Korea Investment & Securities Co., Ltd. stock rating allocation (as of June 30, 2025)**

	BUY	Hold	Underweight (Sell)
Korea Investment & Securities Research Coverage*	77.9%	22.1%	0%
Korea Investment & Securities IB clients**	2.5%	8.8%	0%

Note: * Percentage of companies under coverage with this rating

** Percentage of subject companies within each of the "BUY," "Hold" and "Underweight (Sell)" category for which Korea Investment & Securities Co., Ltd. has provided investment banking services within the previous 12 months.

■ **Guide to Korea Investment & Securities Co., Ltd. sector ratings for the next 12 months**

- Overweight: Recommend increasing the sector's weighting in the portfolio compared to its respective weighting in the Kospi (Kosdaq) based on market capitalization.
- Neutral: Recommend maintaining the sector's weighting in the portfolio in line with its respective weighting in the Kospi (Kosdaq) based on market capitalization.
- Underweight: Recommend reducing the sector's weighting in the portfolio compared to its respective weighting in the Kospi (Kosdaq) based on market capitalization.

■ **Analyst Certification**

I/We, as the research analyst/analysts who prepared this report, do hereby certify that the views expressed in this research report accurately reflect my/our personal views about the subject securities and issuers discussed in this report. I/We do hereby also certify that no part of my/our compensation was, is, or will be directly or indirectly related to the specific recommendations or views contained in this research report.

■ **Important compliance notice**

There is no actual, material conflict of interest of the research analyst or Korea Investment & Securities Co., Ltd., or its affiliates known at the time of publication of the research report or at the time of the public appearance.

Korea Investment & Securities Co., Ltd., or its affiliates has not managed or co-managed a public offering of securities for the companies mentioned in this report in the past 12 months;

Korea Investment & Securities Co., Ltd., or its affiliates has not received compensation for investment banking services from the companies mentioned in this report in the past 12 months; Korea Investment & Securities Co., Ltd., or its affiliates does not expect to receive or intend to seek compensation for investment banking services from the companies mentioned in this report in the next 3 months.

Korea Investment & Securities Co., Ltd., or its affiliates was not making a market in securities of the companies mentioned in this report at the time that the research report was published.

Korea Investment & Securities Co., Ltd. does not own over 1% of shares of the companies mentioned in this report as of July 14, 2025.

Korea Investment & Securities Co., Ltd. has not provided this report to various third parties.

Neither the analyst/analysts who prepared this report nor their associates own any shares of the company/companies mentioned in this report as of July 14, 2025.

Prepared by: Kyungtae Kang

Global Disclaimer

■ General

This research report and marketing materials for Korean securities are originally prepared and issued by the Research Center of Korea Investment & Securities Co., Ltd., an organization licensed with the Financial Supervisory Service of South Korea. The analyst(s) who participated in preparing and issuing this research report and marketing materials is/are licensed and regulated by the Korea Financial Investment Association in Korea only. This report and marketing materials are copyrighted and may not be copied, redistributed, forwarded or altered in any way without the consent of Korea Investment & Securities Co., Ltd.

This research report and marketing materials are for information purposes only. They are not and should not be construed as an offer or solicitation of an offer to purchase or sell any securities or other financial instruments or to participate in any trading strategy. This research report and marketing materials do not provide individually tailored investment advice. This research report and marketing materials do not take into account individual investor circumstances, objectives or needs, and are not intended as recommendations of particular securities, financial instruments or strategies to any particular investor. The securities and other financial instruments discussed in this research report and marketing materials may not be suitable for all investors. The recipient of this research report and marketing materials must make their own independent decisions regarding any securities or financial instruments mentioned herein and investors should seek the advice of a financial adviser. Korea Investment & Securities Co., Ltd. does not undertake that investors will obtain any profits, nor will it share with investors any investment profits. Korea Investment & Securities Co., Ltd., its affiliates, or their affiliates and directors, officers, employees or agents of each of them disclaim any and all responsibility or liability whatsoever for any loss (director consequential) or damage arising out of the use of all or any part of this report or its contents or otherwise arising in connection therewith. Information and opinions contained herein are subject to change without notice and may differ or be contrary to opinions expressed by other business areas or Korea Investment & Securities Co., Ltd. The final investment decision is based on the client's judgment, and this research report and marketing materials cannot be used as evidence in any legal dispute related to investment decisions.

■ Country-specific disclaimer

United States: This report is distributed in the U.S. by Korea Investment & Securities America, Inc., a member of FINRA/SIPC, and is only intended for major U.S. institutional investors as defined in Rule 15a-6(a)(2) under the U.S. Securities Exchange Act of 1934. All U.S. persons that receive this document by their acceptance thereof represent and warrant that they are a major U.S. institutional investor and have not received this report under any express or implied understanding that they will direct commission income to Korea Investment & Securities, Co., Ltd. or its affiliates. Pursuant to Rule 15a-6(a)(3), any U.S. recipient of this document wishing to effect a transaction in any securities discussed herein should contact and place orders with Korea Investment & Securities America, Inc., which accepts responsibility for the contents of this report in the U.S. The securities described in this report may not have been registered under the U.S. Securities Act of 1933, as amended, and, in such case, may not be offered or sold in the U.S. or to U.S. person absent registration or an applicable exemption from the registration requirement.

United Kingdom: This report is not an invitation nor is it intended to be an inducement to engage in investment activity for the purpose of section 21 of the Financial Services and Markets Act 2000 of the United Kingdom ("FSMA"). To the extent that this report does constitute such an invitation or inducement, it is directed only at (i) persons who are investment professionals within the meaning of Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended) of the United Kingdom (the "Financial Promotion Order"); (ii) persons who fall within Articles 49(2)(a) to (d) ("high net worth companies, unincorporated associations etc.") of the Financial Promotion Order; and (iii) any other persons to whom this report can, for the purposes of section 21 of FSMA, otherwise lawfully be made (all such persons together being referred to as "relevant persons"). Any investment or investment activity to which this report relates is available only to relevant persons and will be engaged in only with relevant persons. Persons who are not relevant persons must not act or rely on this report.

Hong Kong: This research report and marketing materials may be distributed in Hong Kong to institutional clients by Korea Investment & Securities Asia Limited (KISA), a Hong Kong representative subsidiary of Korea Investment & Securities Co., Ltd., and may not otherwise be distributed to any other party. KISA provides equity sales service to institutional clients in Hong Kong for Korean securities under its sole discretion, and is thus solely responsible for provision of the aforementioned equity selling activities in Hong Kong. All requests by and correspondence with Hong Kong investors involving securities discussed in this report and marketing materials must be effected through KISA, which is registered with The Securities & Futures Commission (SFC) of Hong Kong. Korea Investment & Securities Co., Ltd. is not a registered financial institution under Hong Kong's SFC.

Singapore: This report is provided pursuant to the financial advisory licensing exemption under Regulation 27(1)(e) of the Financial Advisers Regulation of Singapore and accordingly may only be provided to persons in Singapore who are "institutional investors" as defined in Section 4A of the Securities and Futures Act, Chapter 289 of Singapore. This report is intended only for the person to whom Korea Investment & Securities Co., Ltd. has provided this report and such person may not send, forward or transmit in any way this report or any copy of this report to any other person. Please contact Korea Investment & Securities Singapore Pte Ltd in respect of any matters arising from, or in connection with, the analysis or report (Contact Number: 65 6501 5600).

Copyright © 2025 Korea Investment & Securities Co., Ltd. All rights reserved. No part of this report may be reproduced or distributed in any manner without permission of Korea Investment & Securities Co., Ltd.